Assignment - 2

① $$V(x,y) = \sum_{n=1}^{\infty} c_n \cosh\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi y}{a}\right)$$

$$V_0(\pm b, y) = \sum_{n=1}^{\infty} c_n \cosh\left(\frac{n\pi b}{a}\right) \sin\left(\frac{n\pi y}{a}\right) = V_0$$

Show ~~Show~~ that :

$$c_n = \begin{cases} 0 & ; \text{ if } n \text{ is even} \\ \dfrac{4V_0}{\cosh\left(\frac{n\pi b}{a}\right) n\pi} & ; \text{ if } n \text{ is odd} \end{cases}$$

Ans:

$$V_0 = \sum_{n=1}^{\infty} c_n \cosh\left(\frac{n\pi b}{a}\right) \sin\left(\frac{n\pi y}{a}\right)$$

$$\int_0^a V_0 \sin\left(\frac{m\pi y}{a}\right) dy = \sum_{n=1}^{\infty} c_n \cosh\left(\frac{n\pi b}{a}\right) \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy$$

$\Rightarrow$ Now, $$\int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy$$

$$= \frac{1}{2} \int_0^a \cos\left(\overset{n-m}{\cancel{m-n}}\right) \frac{\pi y}{a} - \cos(m+n) \frac{\pi y}{a} \, dy$$

$$= \frac{1}{2} \left[\frac{\sin(n-m)\frac{\pi y}{a}}{(n-m)\frac{\pi y}{a}} - \frac{\sin(m+n)\frac{\pi y}{a}}{(m+n)\frac{\pi y}{a}} \right]_0^a$$

$$= 0$$

$$\int_0^a \sin^2\left(\frac{n\pi y}{a}\right) dy = \frac{1}{2} \int_0^a 1 - \cos\left(\frac{2n\pi y}{a}\right) dy$$

$$= \frac{1}{2} \left[y - \sin\left(\frac{2n\pi y}{a}\right) \cdot \frac{a}{2n\pi} \right]_0^a$$

$$= \frac{a}{2}$$

$$\therefore \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy = \begin{cases} 0 & ; m \neq n \\ \frac{a}{2} & ; m = n \end{cases}$$

$\therefore$ RHS : $$\sum_{n=1}^{\infty} c_n \cosh\left(\frac{n\pi b}{a}\right) \int_0^a (\quad)$$

$$= \frac{a c_m}{2} \cosh\left(\frac{m\pi b}{a}\right)$$

LHS: $\int_0^a V_0 \sin\left(\frac{m\pi y}{a}\right) dy = V_0 \left[-\cos\left(\frac{m\pi y}{a}\right)\right]_0^a \times \frac{a}{m\pi}$

$$= \frac{V_0 a}{m\pi}(1-\cos m\pi)$$

$$\therefore \frac{\cancel{a} c_m}{2} \cosh\left(\frac{m\pi b}{a}\right) = \frac{V_0 \cancel{a}}{m\pi}(1-\cos m\pi)$$

$$\Rightarrow c_m = \frac{2V_0}{m\pi}(1-\cos m\pi) \times \frac{1}{\cosh\left(\frac{m\pi b}{a}\right)}$$

If $m \to$ even; $\cos m\pi = 1 \Rightarrow c_m = 0$.

If $m \to$ odd; $\cos m\pi = -1 \Rightarrow c_m = \frac{4V_0}{m\pi} \frac{1}{\cosh\left(\frac{m\pi b}{a}\right)}$

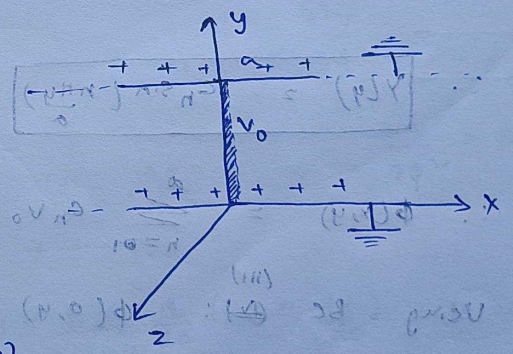
(Proved)

$$\therefore V(x,y) = \sum_{n=1}^{\infty} \frac{4V_0}{n\pi} \frac{\cosh\left(\frac{n\pi x}{a}\right)}{\cosh\left(\frac{n\pi b}{a}\right)} \sin\left(\frac{n\pi y}{a}\right) \quad \text{(Ans)}$$

② 2 ∞ grounded plates lie $\parallel$ to the xz plane one @ $y=0$ & one @ $y=a$. The left end @ $x=0$ is closed off with an ∞ metal strip, insulated from the 2 plates & maintained @ a potential V_0. Find the potential in the region b/w the 2 plates (Assume $V \to 0$ as $x \to \infty$).

Ans: Domain:

$0 < y < a$; $x > 0$



Let the potential b/w the 2 plates be $\phi(x,y)$.

$\therefore$ Boundary conditions (BCs)

(i) $\phi(x,0) = 0$ (ii) $\phi(x,a) = 0$

(iii) $\phi(0,y) = V_0$ (iv) $\phi(\infty,y) \to 0$

Let $\phi(x,y) = X(x) \cdot Y(y)$

Laplace's eqn: $\nabla^2 \phi = 0 \Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$

$$\frac{\partial^2\phi}{\partial x^2} = X''Y \quad ; \quad \frac{\partial^2\phi}{\partial y^2} = Y''X.$$

$$\therefore \quad X''Y + Y''X = 0 \Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = k^2$$

$\boxed{X}$: $\frac{X''}{X} = k^2 \Rightarrow X'' - k^2 X = 0$

$\therefore \; r^2 - k^2 = 0 \Rightarrow r = \pm k.$

$$X(x) = A_1 e^{-kx} + A_2 e^{kx}$$

BC (IV): $X(\infty) \to 0 \Rightarrow \boxed{A_2 = 0}$

BC (III): $X(0) = V_0 \Rightarrow \boxed{A_1 = V_0}$

$\therefore \; \boxed{X(x) = V_0 e^{-kx}}$

$\boxed{Y}$: $Y'' + k^2 Y = 0$

$r^2 + k^2 = 0 \Rightarrow r = \pm ik.$

$$Y(y) = B_1 e^{-iky} + B_2 e^{iky}$$

$$= c_1 \sin(ky) + c_2 \cos(ky).$$

BC (I): $Y(0) = 0 \Rightarrow \boxed{c_2 = 0}$

BC (II): $Y(a) = 0 \Rightarrow c_1 \sin ka = 0$

$\Rightarrow \boxed{k = \frac{n\pi}{a}} \quad n = 1, 2, \ldots$

$\therefore \; \boxed{Y(y) = c_n \sin\left(\frac{n\pi y}{a}\right)}$

$$\therefore \; \phi(x,y) = \sum_{n=1}^{\infty} c_n V_0 e^{-\frac{n\pi x}{a}} \sin\left(\frac{n\pi y}{a}\right)$$

Using BC (III): $\phi(0,y) = V_0.$

$$\Rightarrow \sum_n c_n \cancel{V_0} \sin\left(\frac{n\pi y}{a}\right) = \cancel{V_0}$$

$$\Rightarrow \sum_n c_n \sin\left(\frac{n\pi y}{a}\right) = 1$$

(from (Q1))

$$\Rightarrow \sum_n c_n \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy = \int_0^a \sin\left(\frac{m\pi y}{a}\right) dy$$

$$\Rightarrow c_m \frac{a}{2} = \left[-\cos \frac{m\pi y}{a}\right]_0^a \cdot \frac{a}{m\pi} = \frac{a}{m\pi}(1 - \cos m\pi)$$

$\Rightarrow \quad c_m = \frac{2}{m\pi}(1 - \cos m\pi)$

If $m \rightarrow$ even : $c_m = 0$.

If $m \rightarrow$ odd : $c_m = \frac{4}{m\pi}$.

$$\therefore \boxed{\phi(x,y) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_0}{n\pi} \sin\left(\frac{n\pi y}{a}\right) e^{-\frac{n\pi x}{a}}} \quad \text{(Ans)}$$